THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 03

Full Syllabus (Chapters 1–14) | Class 8 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** 14	There are $2n = 14$ numbers between n^2 and $(n+1)^2$
2	**(a)** 2	$72 = 2^3 \times 3^2$; one more 2 is needed
3	**(d)** 5^9	Add the exponents: $2 + 7 = 9$
4	**(b)** 222	Repeated division of 26 by 3
5	**(b)** parallelogram	Converse property
6	**(a)** 115°	Adjacent angles are supplementary
7	**(a)** Yes	Digit sum 18 is divisible by 9
8	**(a)** 7	$5 + 2$
9	**(d)** 45	Product of extremes = product of means: $4x = 180$
10	**(d)** ₹ 360	A gets 3/7 of 840
11	**(d)** ₹ 1210	1000 × 1.1 × 1.1
12	**(b)** 20%	Profit or loss on CP
13	**(a)** $a\sqrt{2}$	Diagonal of the original square
14	**(a)** 17 cm	$\sqrt{64 + 225}$
15	**(d)** 1 km	100000 cm = 1 km
16	**(a)** 9	Euler's relation
17	**(a)** 10	All equal
18	**(b)** $10 \times 3 + 7$	Place value
19	**(d)** 50 cm²	$\frac{1}{2}(a+b)h$
20	**(c)** equal area	Same height and equal bases

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	$625 = 5^4$ so $x = 4$ (2).	2
22	They learnt them from the Arab world (2).	2
23	100 (1); the sum of first n odd numbers is n^2 (1).	1+1
24	$24 : 6 = 4 : 1$ (2).	2
25	$7\sqrt{2} \approx 9.9$ cm (2).	2
26	8 days (2).	2
27	$10/6 \approx 1.67$°C per hour (2).	2
28	42 cm² (2).	2

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	$12^2 = 144$, $12^3 = 1728$ (2); difference 1584 (1).	2+1
30	$\sqrt{40^2 + 30^2}$ (2); 50 m (1).	2+1
31	$16 + 4 + 0.25 = 20.25$ m² (3).	3
32	$A = P(1 + r/100)^n = 12000 \times (1 + 6/100)^2 = ₹13,483.20$ (2); CI = ₹1,483.20 (1).	2+1
33	Surface area $= 6 \times 3^2 = 54$ cm² (2); 2 cubes (1).	2+1
34	$4x + 16 = 92$ (1.5); $x = 19$ (1.5).	1½+1½
35	Diagonal 29 m (1.5); saved $41 - 29 = 12$ m (1.5).	1½+1½
	OR: 13 m (3).	3

36	$\frac{1}{2}(30)h = 150$ (2); $h = 10$ cm (1).	2+1
	OR: Area 240 cm² (1.5); height $240/13 \approx 18.5$ cm (1.5).	1½+1½

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) $\frac{2^8 \times 2^3}{2^6} = 2^5 = 32$ (3). (ii) 2^5 (2).	3+2
	OR: (i) $6^3 = 216 = 8 \times 27$ (2). (ii) $(12/6)^3 = 8$ (3).	2+3
38	(i) 12000 then 10800 (3). (ii) 28% (2).	3+2
	OR: (i) $1.1 \times 1.1 = 1.21$ (3). (ii) $1.1 \times 0.9 = 0.99$, a 1% decrease (2).	3+2
39	(i) Both are multiplied by 2 (3). (ii) Mean 5 to 10, median 5 to 10 (2).	3+2
	OR: (i) 42 (2). (ii) For 17 numbers, the median is the 9th value (3).	2+3
40	(i) Half difference 20; height $\sqrt{50^2 - 20^2} \approx 45.8$ m (2). (ii) $\frac{1}{2} \times 200 \times 45.8 \approx 4580$ m² (1). (iii) ≈ ₹ 68700 (2).	2+1+2
	OR: (i) Square 144, rectangle $16 \times 8 = 128$ (3). (ii) Square has the larger area (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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